How-To Library

Informatica

Using PowerExchange ODBC Connections in a Data Archive Retirement Project

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Abstract

You can use PowerExchange ODBC connections to connect to z/OS data sources in an Informatica Data Archive retirement project. This article describes the procedures for using PowerExchange ODBC connections in an archive retirement project.

Supported Versions

- Data Archive 5.3.7 or later
- PowerExchange 10.0 or later

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Overview

This article summarizes the steps that are required to use PowerExchange ODBC connections in a Data Archive retirement project.

A retirement archive project copies a set of tables from a source database to an archive database or optimized file archive. The source database can then be decommissioned. If necessary, the archived data can be restored by using Information Lifecycle Management (ILM) tools.

To retire IMS or VSAM source data on a z/OS system, you can use PowerExchange ODBC connections that are provided with PowerExchange. The Enterprise Data Manager (EDM) and Data Archive use the JDBC-ODBC bridge to connect to the z/OS source data through the PowerExchange ODBC drivers.

This article assumes that you have installed Informatica Data Archive, PowerCenter, and Data Validation Option (DVO) and are familiar with these products. Familiarity with PowerExchange is also recommended.

Implementation Scenario

A data archive retirement project uses PowerExchange ODBC to access IMS and VSAM data sources on z/OS. This implementation includes the following systems and components:

- A z/OS system with the IMS and VSAM data sources.

  On this system, you will install PowerExchange 9.1.0 HotFix 4 or later.
- A Linux, UNIX, or Windows system that has the following type of storage and software:

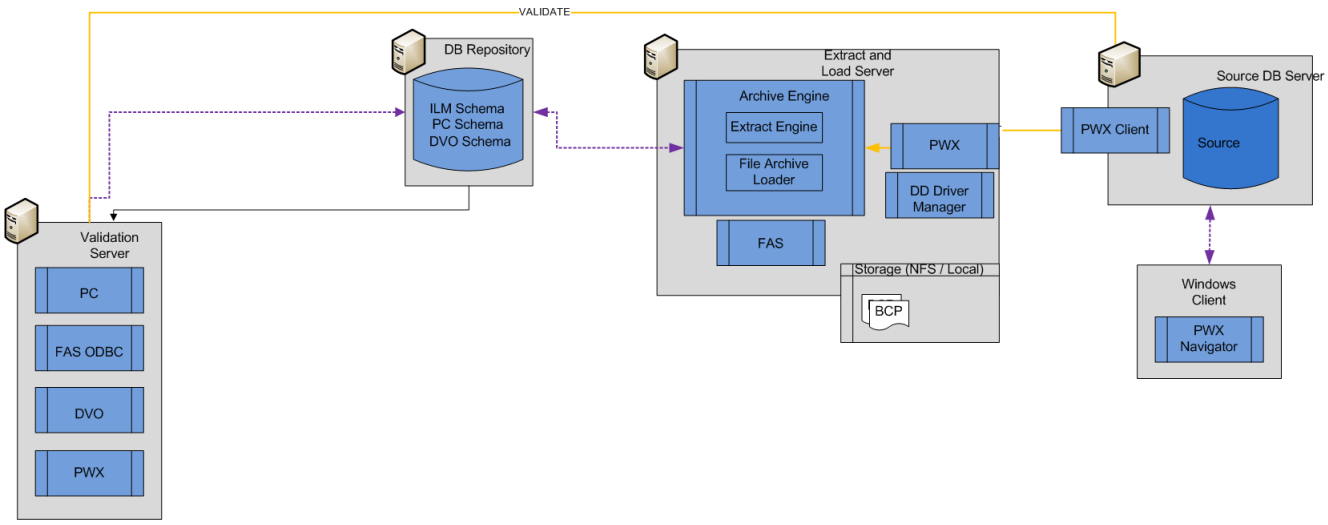
  - Informatica Data Archive engine 5.3.7 or later
  - File Archive Server (FAS)
  - NFS or local storage for the encapsulated BCP files

  On this system, you will install PowerExchange 9.1.0 HotFix 4 or later. You will also copy the DataDirect Driver Manager and define PowerExchange ODBC data sources.
- Database repository for ILM, PowerCenter, and DVO schema.
- Validation server with the PowerCenter Domain, PowerCenter Client, FAS ODBC, and DVO installed.

  On this system, you will install PowerExchange and define PowerExchange ODBC data sources.
- A Microsoft Windows system.

  On this system, you will install PowerExchange 9.1.0 HotFix 4 or later and run the PowerExchange Navigator.

The following figure illustrates the implementation. In this figure, PWX represents PowerExchange, PC represents PowerCenter, and DD represents DataDirect.



Task Flow

To use PowerExchange ODBC connections in a Data Archive retirement project, complete the following steps:

1. Install and configure PowerExchange on z/OS.
2. Install and configure PowerExchange on the Data Archive server.
3. Install and configure PowerExchange on the Validation server.
4. Install and configure PowerExchange on Windows.
5. Define data maps for the z/OS data sources.
6. Configure PowerExchange ODBC data sources on the Data Archive server.
7. Configure PowerExchange ODBC data sources on the Validation server.
8. Import z/OS metadata.
9. Define and run the retirement archive project.
10. Validate the retired data.

Step 1: Install and Configure PowerExchange on z/OS

To install PowerExchange on z/OS, run the MVS Installation Assistant. After you install PowerExchange, edit the DBMOVER configuration file and start the PowerExchange Listener.

Install PowerExchange on z/OS

Install Power Exchange on the z/OS system where the source data for the retirement project is located.

To install PowerExchange on z/OS, run the MVS Installation Assistant. For detailed instructions, see the *PowerExchange Installation Guide*. This guide gives details about how access to the z/OS databases and files is controlled.

Edit the DBMOVER Configuration Member on the z/OS System

Edit the DBMOVER configuration member.

Verify that the member includes a LISTENER statement. The LISTENER statement defines the TCP/IP port on which a named PowerExchange Listener process listens for work requests. The default DBMOVER member includes the required LISTENER statement.

In addition to the LISTENER statement, you can include the following statements in the DBMOVER configuration member to minimize data map I/O:

```
DMXCACHE_MAX_MEMORY_MB=20
DM_SUBTASK=R
```

When Data Archive accesses PowerExchange data maps through the PowerExchange Listener, every data map in the DATAMAPS KSDS is opened and processed. The DMXCACHE_MAX_MEMORY_MB and DM_SUBTASK statements configure the PowerExchange Listener on z/OS to use data maps caching and a data maps subtask in read-only mode.

Note: To further reduce data map I/O, limit the number of data maps in the DATAMAPS file. For best results, limit the number of data maps to 100.

For more information about the DBMOVER configuration member, see the *PowerExchange Reference Manual*.

Configure the PowerExchange Listener JCL on z/OS

Configure the PowerExchange Listener JCL before starting the PowerExchange Listener the first time. PowerExchange provides PowerExchange Listener JCL in the following members of the RUNLIB library:

- PWXLSTNR runs PowerExchange as a started task.
- STARTLST runs PowerExchange as a batch job.

Note: When you ran the XIZZZ998 job during installation, the PWXLSTNR member was copied to the PROCLIB library.

1. Verify that the STEPLIB DD concatenation in the PowerExchange Listener JCL includes the load libraries for your source DBMSs.

   Usually, the PowerExchange LOADLIB library occurs before the data-source load libraries in the concatenation.

   The STEPLIB concatenation in the STARTLST member includes the load libraries for all data sources in a proper order.
2. Verify that the PARM statement specifies the node name that is defined in the LISTENER statement of the DBMOVER configuration file.
3. In the REGION parameter of the EXEC statement, enter a region size. The region size indicates the amount of storage to allocate for running the PowerExchange Listener.

   The sample JCL specifies a region size of 128 MB in the EXEC statement.

   Informatica recommends a region size of 0M, which indicates no storage limit.

   If you omit the REGION parameter, the system default applies.
4. Verify that the TIME parameter in the EXEC statement is set to NOLIMIT. The NOLIMIT setting enables the PowerExchange Listener job to run without a time limit and prevents timeout-related abends that might occur with abend code S322.
5. If the SECURITY statement in the DBMOVER configuration file is set to 1 or 2, APF-authorize the PowerExchange LOADLIB library and any other load libraries in the STEPLIB DD concatenation of the PowerExchange Listener JCL.

For more information, see the *PowerExchange Bulk Data Movement Guide*.

Start the PowerExchange Listener

To start the PowerExchange Listener, issue the MVS START command with the name of the started task. For example:

```
START PWXLSTNR
```

You can also run the PowerExchange Listener as a batch job. However, because it is a long-running task, using an MVS started task is more appropriate.

Step 2: Install and Configure PowerExchange on the Data Archive Server

Install and configure PowerExchange on the same server as Informatica Data Archive.

Install PowerExchange on the Data Archive Server

On the Data Archive server, install PowerExchange 9.1.0 HotFix 4 or later. Use the PowerExchange installation instructions for the operating system type and bit-level. If the Data Archive server runs on a 64-bit machine, use the PowerExchange 64-bit installation executable or tar file. If the Data Archive server runs on a 32-bit machine, use the PowerExchange 32-bit executable or tar file.

If you are installing on Windows, install the PowerExchange ODBC drivers in the Windows registry:

- On Windows 32-bit systems other than Vista, enter the following command at the command prompt:

  ```
  dtlodbci add
  ```

- On Windows Vista, update the dtlodbci.exe file properties to select **Run this program as an administrator**, and then enter the following command at the command prompt:

  ```
  dtlodbci add
  ```

- On 64-bit Windows, update the registry manually.

For detailed installation instructions, see the *PowerExchange Installation Guide*.

Edit the DBMOVER Configuration File on the Data Archive Server

Edit the DBMOVER configuration file.

Include a NODE statement to define the TCP/IP host name and port that PowerExchange uses to contact the PowerExchange Listener on the machine where the source data for the retirement project is located. Use the following format:

```
NODE=({node_name|node1}
      ,TCPIP
      ,host_name
     ,{port|2480}
    [,{send_bufsize|65536}]
    [,{receive_bufsize|65536}]
    [,{send_size|4096}]
    [,{receive_size|4096}]
    [,receive_timeout]
    [,{SSL|ZOSSSL}]
    [,service_name]
)
```

For more information, see the *PowerExchange Reference Manual*.

Configure Environment Variables on Linux or UNIX

If the Data Archive server is running on Linux or UNIX, edit the path and library path environment variables on the Linux or UNIX system to point to the directories where the files and libraries reside.

Also, set the PWX_HOME environment variable so that PowerExchange can locate various run-time components.

The following table lists the Linux and UNIX environment variables that you need to set:

Variable	Platform	Example
PWX_HOME	All	PWX_HOME=/usr/pwxuser/v910
PATH	All	PATH=/usr/pwxuser/v910
LIBPATH	AIX	LIBPATH=/usr/pwxuser/v910
LD_LIBRARY_PATH	Sun	LD_LIBRARY_PATH=/usr/pwxuser/v910
LD_LIBRARY_PATH	Linux	LD_LIBRARY_PATH=/usr/pwxuser/v910
SHLIB_PATH	HP-UX 10/11	SHLIB_PATH=/usr/pwxuser/v910

To verify that the environment variables are correct, log off and log in again.

Note: After you create ODBC data sources on UNIX, you must update the library path and ODBCINI environment variables. For more information, see "Configure ODBC Data Sources on Linux or UNIX" on page 9.

Step 3: Install and Configure PowerExchange on the Validation Server

Install and configure PowerExchange on the Validation server, as follows:

- To install PowerExchange on the Validation server, use the same procedure as for installing PowerExchange on the Data Archive server. See " Step 2: Install and Configure PowerExchange on the Data Archive Server" on page 6.
- Edit the DBMOVER configuration file.

  Include a NODE statement to define the TCP/IP host name and port that PowerExchange uses to contact the PowerExchange Listener on the machine where the source data for the retirement project is located.

  For more information, see the *PowerExchange Reference Manual*.
- On Linux or UNIX, set the PWX_HOME environment variable so that PowerExchange can locate various run-time components. For example:

```
PWX_HOME=/usr/pwxuser/v910
```

Step 4: Install and Configure PowerExchange on Windows

On the Windows system on which you will run the PowerExchange Navigator, install Power Exchange and edit the DBMOVER configuration file.

Install PowerExchange on Windows

On the Windows machine, run the PowerExchange 32-bit executable file to install PowerExchange. You must use the Windows 32-bit executable to install the PowerExchange Navigator, even on a Windows 64-bit machine.

For detailed installation instructions, see the *PowerExchange Installation Guide*.

Edit the DBMOVER Configuration File on the Windows System

Edit the DBMOVER configuration file.

Include a NODE statement to define the TCP/IP host name and port that PowerExchange uses to contact the PowerExchange Listener on the machine where the source data for the retirement project is located.

For more information, see the *PowerExchange Reference Manual*.

Step 5: Create PowerExchange Data Maps

For each IMS and VSAM data source that you want to retire, create a PowerExchange data map.

Before you create the data maps, select to use a two-tier naming convention for the data maps. To use a two-tier naming convention, click **Options > Preferences**, and on the **Data Maps** tab in the **Preferences** dialog box, select the **Use 2-tier names** option.

1. On the **Resources** tab in the **Resource Explorer**, click **Add** > **Data Map**.
2. In the **Name** dialog box, enter the following information:

Option	Description
Schema Name	Enter a user-defined schema name. PowerExchange uses this name as the first part of the data map name, as follows: `schema_name.data_map_name` The name must consist of alphanumeric characters and begin with a letter. Maximum length is 10 characters.
Data Map Name	Enter a user-defined map name. PowerExchange uses this name as the second part of the data map name, as follows: `schema_name.data_map_name` The name must consist of alphanumeric characters and begin with a letter. Maximum length is 10 characters.
Access Method	Select one of the following access methods: - **DL/1 BATCH**. IMS database on z/OS. - **ESDS**. VSAM Entry Sequenced Data Set (ESDS) on z/OS. - **IMS ODBA**. IMS database accessed through a PowerExchange Listener on z/OS. - **KSDS**. VSAM Key Sequenced Data Set (KSDS) on z/OS. - **RRDS**. VSAM Relative Record Data Set (RRDS) on z/OS.
Import Record Definitions	Select this option to import a copybook into the data map. Clear this option to add only the data map. By default, this option is selected.

3. Enter any additional information that is required for the access method that you selected.
4. If you selected the **Import Records Definitions** option, click **Next**.

   If you cleared this option, click **Finish**.
5. If you selected the **Import Records Definitions** option, provide the required information in the dialog boxes.

For more information about creating data maps, see the *PowerExchange Navigator User Guide*.

Step 6: Configure PowerExchange ODBC Data Sources on the Data Archive Server

Configure PowerExchange ODBC data sources on the Linux, UNIX, or Windows system where the Data Archive server and PowerExchange are installed.

Configure ODBC Data Sources on Linux or UNIX

Use this procedure to configure ODBC data sources on Linux or UNIX.

1. Copy the DataDirect Driver Manager from the computer where PowerCenter and DVO are installed to the server where Power Exchange and the Data Archive engine are installed.

   To copy the DataDirect Driver Manager, copy the following directory to the system with the Data Archive engine:

   PowerCenter_installation_directory/ODBC6.1

   Tar the entire directory, copy it using FTP, and then untar the directory on the target system.
2. Add the ODBC6.1/lib directory to one of the following environment variables:
   - LD_LIBRARY_PATH on Sun or Linux
   - SHLIB_PATH on HP-UX
   - LIBPATH on AIX
3. Create the odbc.ini file and set the ODBCINI environment variable to point to this file.

   In the odbc.ini file, include the following line for each data source:

   ```
   Compatibility=ILMBRIDGE
   ```

   The following example shows a section of an odbci.ini file for one data source:

   ```
   [nrdb2_mhz8902]
   Driver=/pwx_installation_directory/source/libdtlodbc.so
   Description=dBase
   Database=
   DBType=nrdb2
   Location=zos_listener_name
   DBQual1=
   DBQual2=
   MaxRows=0
   Compress=N
   Encrypt=N
   ConfirmWrite=N
   PaceSize=0
   InterpretASRows=N
   BulkLoad=N
   DeleteTempFiles=0
   LoadOptions=0
   JCLTemplate=
   CTLTemplate=
   ModeType=0
   ModeTime=0
   Time=0
   Space=
   pace=0
   SecSpace=0
   Compatibility=ILMBRIDGE
   ```
4. For HP-UX systems, change the driver name in the odbc.ini file from libdtlodbc.so to libdtlodbc.sl.

For more information, see the *PowerExchange Reference Manual*.

Configure ODBC Data Sources on Windows

Use this procedure to configure ODBC data sources on Windows.

Before you configure ODBC data sources on Windows, ensure that the PowerExchange ODBC drivers have been added to the Windows registry.

1. Copy the DataDirect Driver Manager from the computer where PowerCenter and DVO are installed to the server where Power Exchange and the Data Archive engine are installed.

   To copy the DataDirect Driver Manager, copy the following directory to the system with the Data Archive engine:

   ```
   PowerCenter_installation_directory/ODBC6.1
   ```

2. To start the 32-bit **ODBC Data Source Administrator** wizard, perform one of the following actions:
   - On a Windows 32-bit system, double-click **Administrative Tools** in the **Control Panel**. In the **Administrative Tools** window, double-click **Data Sources (ODBC)**.
   - On a Windows 64-bit system, enter the following command at the command prompt:

     ```
     %windir%\SysWOW64\odbcad32.exe
     ```

   The **ODBC Data Source Administrator** wizard appears.
3. On the **System DSN** tab, click **Add**.
4. In the **Create New Data Source** dialog box, select the **Informatica PowerExchange** driver from the list of available drivers and click **Finish**.

   The **PowerExchange Data Source** wizard appears.
5. On the **PowerExchange Data Source** tab, enter the following information:
   - At **Location**, enter the name of the PowerExchange Listener, as defined in the NODE statement in the DBMOVER configuration file.
   - At **Type**, select **NRDB2**.
   - At **Local Codepage**, select **UTF-8 encoding of Unicode**.
   - Depending on the data source type that you select, enter values for other available properties.
6. On the **General** tab, enter the following values at **Integration Mask**:

   ```
   ILMBRIDGE,CPOOL
   ```

   ILMBRIDGE sets a number of options that are required by the JDBC-ODBC bridge. CPOOL enables PowerExchange connection pooling.

   For all data source types, you can optionally enter additional information.
7. Click **OK**.

   The ODBC data source appears in the **System Data Sources** list on the **System DSN** tab in the **ODBC Data Source Administrator** wizard.
8. Click **OK**.

For more information, see the *PowerExchange Reference Manual*.

Step 7: Configure PowerExchange ODBC Data Sources on the Validation Server

Configure PowerExchange ODBC data sources on the Validation server. Use the same procedures that you used to configure the data sources on the Data Archive server. For more information, see " Step 6: Configure PowerExchange ODBC Data Sources on the Data Archive Server" on page 9.

Step 8: Import z/OS Metadata

Use the Enterprise Data Manager (EDM) to import metadata for the z/OS data sources. Perform this procedure for each data source.

1. To start the Enterprise Data Manager, at the Information Lifecycle Management main window, select **Accelerators > Enterprise Data Manager**.
2. To create an application version for the data source, right-click **Custom Apps Only** in the navigator pane, select **New Application Version**, and enter an application name and description.
3. To import PowerExchange metadata for the application version that you created, complete the following tasks:
   a. Highlight the object and click **File > Mine Database**.

      The **Connect to Mined Database** dialog box appears.

   b. If you have already defined a connection, select it, and enter your z/OS user name and password. The remaining connection information is filled in automatically.

      If you are defining a new connection, enter the following information:

      - At **Database Type**, select **CUSTOM_JDBC**.
      - At **JDBC URL**, enter `jdbc:odbc:`*`dsn_name`*, where *dsn_name* matches the name in the odbc.ini file.
      - At **Driver**, enter `sun.jdbc.odbc.JdbcOdbcDriver`.
      - Enter your z/OS user name and password.

   c. In the **Database Mining** wizard, select the schema and list of tables for the data source.
   d. At the prompt to mine child tables of the selected tables, click **No**.

      Note: Because the PowerExchange NRDB2 access method does not support foreign key relationships, you cannot mine child tables. By clicking **No**, you ensure that the PowerExchange Listener on z/OS does not spawn multiple subtasks that would return empty result sets.

      If you enabled the ILMBRIDGE or NOFKEYS Compatibility option for the PowerExchange ODBC connection, the PowerExchange Listener does not spawn multiple empty subtasks, regardless of what you select for this option.

      The EDM begins importing the metadata and displays the percentage complete in the **Database Mining Progress** window.
4. To create and configure an entity object for the application version, complete the following tasks:
   a. Create an entity object and interim table.
   b. Insert tables into the entity.
   c. To avoid validation warnings, mark the first table as the driving table.
   d. Add tables to the interim table.
   e. Add Insert and Delete statements for each table.

      For retirement projects, you typically select a true statement such as **1=1** for the Insert statement and a false statement such as **1=2** for the Delete statement.

   f. Set default columns information for the interim table. For the From clause, enter the name of the driving table. For the Where clause, enter **1=1**.
   g. Delete unneeded entity steps. For retirement projects, you can delete all steps except the Insert into Archive Tables steps.
5. Click **Save**, and close the EDM.

For more information about importing metadata into Data Archive, see the *Data Archive Enterprise Data Manager* guide.

Step 9: Define and Run the Retirement Project

In this step, you define and run the retirement archive project. Running the project copies the nonrelational data to a target database or optimized file archive.

Define the Retirement Archive Project

To define a retirement archive project, specify the source and target connections, and define the retirement project application details.

1. To define a source connection, select **Administration > New Archive Source** from the **Information Life Cycle** main window, and enter the following information:
   - At **Connection Type**, select **JDBC-ODBC_BRIDGE 1.0**.
   - Enter the **Data Source Name** as defined in the odbc.ini file.
   - For the **Admin**, **Application**, and **Staging** login names and passwords, enter a valid z/OS user ID and password.
   - At **JDBC Fetch Size**, enter 0.
   - Clear the **Use Staging** check box.
2. Define the target connection, if you have not already defined it.
3. Define the retirement project:
   a. Click **Workbench > Manage Retirement Projects**, and then click **New Retirement Project**.
   b. In the **New/Edit Retirement Project** window, complete the application details.
   c. Select the data source and target connection that you just defined.
   d. Click **Add Entity**, and select the entity for the data source that you are retiring.
   e. Click **Next** until the **Manage Execution** window is displayed. Then select **Publish** to save the retirement project.

When you click the **Publish** button, Data Archive lists the project in the **Manage Retirement Projects** page. By publishing the project, you indicate that the project is ready for scheduling.

For more information, see the *Data Archive User Guide*.

Run the Retirement Project

Schedule or run the retirement project that you published in the previous task.

1. Select **Workbench > Manage Retirement Projects**, and select the project.
2. Complete the tasks to schedule the retirement project and confirm the retirement run.

For more information, see the *Data Archive User Guide*.

Step 10: Validate the Retired Data

Use DVO to verify that the data that was copied to the data archive matches the source data on z/OS.

For more information, see the *PowerCenter Data Validation Option Installation and User Guide*.

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